

Draft - expert review pending

This unit is complete and passes the platform's automated checks, but no reviewer has read it yet. Treat its sources and claims with care.

Draft translation - not yet human-reviewed / مسوده ترجمه

Cognitive Processes in Learning

Unit 3 asked what learning is. This unit asks what the learner's mind is doing while it happens. The question is practical, not philosophical: a lesson that demands what attention cannot hold, or teaches in an order memory cannot keep, fails for reasons no amount of effort fixes. And a lesson that works *with* the mind's systems - one thing at a time, retrieval practised, steps made visible - succeeds for reasons that look like talent and are not.

Three topics: the gateway (attention and perception), the workbench (memory and the information-processing model), and the workshop (thinking, problem-solving and reasoning).

Unit learning outcomes

By the end of this unit you should be able to:

- Explain **attention** as the gateway to learning and describe what captures, divides and exhausts it in a real classroom.
- Explain **perception** as active interpretation, and what that means for how pupils read a lesson's materials.
- Describe the **information-processing model**: sensory memory, working memory and long-term memory, and the jobs of encoding, storage and retrieval.
- Explain why **forgetting** happens - decay, interference and retrieval failure - and what the evidence says about study strategies that work.

- Describe **thinking and concept formation**, **problem-solving** strategies and their obstacles, and **reasoning** in its inductive and deductive forms.
- Design and justify classroom activities that work **with** attention and memory rather than against them.

Prerequisite knowledge

Units 1-3 (the three-part scope; development, especially the cognitive domain; the cognitive turn in learning theories, which this unit deepens). Unit 5's intelligence material builds on this unit's thinking topics.

In this unit

1. Topic 4.1 - Attention and perception
2. Topic 4.2 - Memory and information processing
3. Topic 4.3 - Thinking, problem-solving and reasoning

How to use this unit

- **Work through the topics in order.** Topic 4.2 assumes the gateway is understood; Topic 4.3 assumes the workbench.
- **Test the claims on yourself.** Every topic's demonstrations work on the reader: try the span task, the gorilla video, the anagram before you read its solution. First-hand failure is the fastest teacher of these limits.
- **Keep one lesson in mind.** The unit assessment asks you to redesign a study or revision plan against the memory model, so the notes you keep on your own study habits are the raw material.
- After the three topics, the **unit assessment** page carries a full question bank with answers and marking guidance.

Tip: select any text on this page first to comment on that specific passage - or leave nothing selected for general feedback.

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